Chapter 10
Relations

The Mechanics of Proof | CSC302 lecture notes

Based on Heather Macbeth's textbook

What we will learn

- Test reflexivity, symmetry, antisymmetry, and transitivity separately.
- Give the exact witnesses needed to refute each property.
- Prove that a relation is an equivalence relation.
- Define equivalence classes and prove why classes agree or are disjoint.
- Recognize equivalence relations on numbers, pairs, sets, and types.

A relation is a predicate with two inputs

A relation r on X assigns a proposition $r(x, y)$ to each pair $x, y \in X$. We write $x \sim y$ for $r(x, y)$. In Lean its type is `X -> X -> Prop`.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 10 introduction; Sections 10.1–10.2.

10.1: Four properties, four different statements

Fix a relation $\sim$ on a type X. All displayed variables range over X.

Property	Definition
Reflexive	$\forall x, \quad x \sim x.$
Symmetric	$\forall x, y, \quad x \sim y \implies y \sim x.$
Antisymmetric	$\forall x, y, \quad (x \sim y \ \wedge \ y \sim x) \implies x = y.$
Transitive	$\forall x, y, z, \quad (x \sim y \ \wedge \ y \sim z) \implies x \sim z.$

Antisymmetry permits $x \sim x$. It prohibits two-way relatedness between *distinct* elements.

Symmetry and antisymmetry are compatible: equality on $\mathbb{R}$ has both properties.

Source: Heather Macbeth, *The Mechanics of Proof*, Definitions in Example 10.1.1; Example 10.1.2.

Counterexamples must match the negated quantifiers

To disprove	Exhibit
Reflexivity	An x for which $x \not\sim x$.
Symmetry	x, y such that $x \sim y$ but $y \not\sim x$.
Antisymmetry	Distinct x, y with both $x \sim y$ and $y \sim x$.
Transitivity	x, y, z with $x \sim y$ and $y \sim z$, but $x \not\sim z$.

Lean pattern. Unfold the property, use `push_neg`, and provide the witnesses with `use`. Then verify each required fact.

For transitivity, just finding an unrelated pair proves nothing. The two premises must hold for the same middle element y.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 10.1.1–10.1.4.

Divisibility on the natural numbers

Let $a \sim b$ mean $a \mid b$, with $a, b \in \mathbb{N}$. Recall: $a \mid b$ means $b = ak$ for some $k \in \mathbb{N}$.

- **Reflexive.** For every a, $a = a \cdot 1$, so $a \mid a$.
- **Not symmetric.** $1 \mid 2$, but $2 \nmid 1$. Indeed, every natural multiple of 2 is either 0 or at least 2.
- **Transitive.** Assume $a \mid b$ and $b \mid c$. Write $b = ak$ and $c = bl$ with $k, l \in \mathbb{N}$. Then

$$c = bl = (ak)l = a(kl).$$

  Since $kl \in \mathbb{N}$, this is a witness for $a \mid c$.

Antisymmetry also holds, but requires attention to the value 0.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.1.1.

Antisymmetry of divisibility: Handle zero before comparing sizes

Suppose $a, b \in \mathbb{N}$ with $a \mid b$ and $b \mid a$. We must prove $a = b$.

1. If $a = 0$, the equation $b = ak$ from $a \mid b$ gives $b = 0$. If $b = 0$, the equation $a = bl$ gives $a = 0$.
2. Otherwise $a, b > 0$. Write $b = ak$ and $a = bl$ with $k, l \in \mathbb{N}$. The positivity of b forces $k \geq 1$, and the positivity of a forces $l \geq 1$.
3. Therefore $b = ak \geq a$ and $a = bl \geq b$, so $a = b$.

The domain matters

On $\mathbb{Z}$, divisibility is not antisymmetric: $1 \mid -1$ and $-1 \mid 1$, but $1 \neq -1$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.1.1; integer contrast added for comparison.

Lean: A transitivity proof combines two witnesses

```
def divides (a b : Nat) : Prop :=
  Exists (fun k : Nat => b = a * k)

example : Transitive divides := by
  intro a b c hab hbc
  cases hab with
  | intro k hk =>
      cases hbc with
      | intro l hl =>
          use k * l
          calc c = b * l := hl
            _ = (a * k) * l := by rw [hk]
            _ = a * (k * l) := by ring
```

The middle value b connects the equations. The new witness is the product kl.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.1.1; textbook-environment demonstration.

Being close is reflexive and symmetric, but need not be transitive

On $\mathbb{R}$, define $x \sim y$ by $(x-y)^2 \leq 1$.

- **Reflexive:** $(x-x)^2 = 0 \leq 1$.
- **Symmetric:** $(y-x)^2 = (x-y)^2$, so the inequality survives reversal.
- **Not antisymmetric:** $0 \sim 1$ and $1 \sim 0$ since both squared differences are 1; yet $0 \neq 1$.
- **Not transitive:** $0 \sim 1$ and $1 \sim 2$, but $(0-2)^2 = 4 > 1$, so $0 \not\sim 2$.

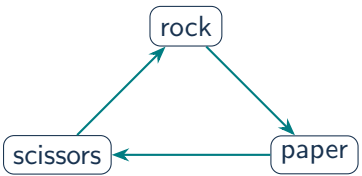


Each short step is allowed; the combined step is not.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.1.3; new counterexample choices and diagram.

A finite directed cycle is antisymmetric

Let $x \prec y$ mean "x loses to y" for rock, paper, and scissors. The diagram lists all related pairs.



- Not reflexive: there is no loop at rock.
- Not symmetric: rock $\prec$ paper, but paper $\not\prec$ rock.
- Antisymmetric: no pair has arrows in both directions, so its two-premise implication always holds.
- Not transitive: rock $\prec$ paper and paper $\prec$ scissors, but rock $\not\prec$ scissors.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.1.4; original drawing.

Practice 10.1: Prove or refute each property

For each relation, decide whether it is reflexive, symmetric, antisymmetric, and transitive.

1. On $\mathcal{P}(\mathbb{N})$, define $U \sim V$ by $U \subseteq V$. Your variables are now sets of natural numbers.
2. On $\mathbb{R}^2$, define

$$(x_1, y_1) \preceq (x_2, y_2) \quad \iff \quad x_1 \leq x_2 \ \wedge \ y_1 \leq y_2.$$

   Your variables are now ordered pairs of real numbers.

Each positive answer needs a universal proof; each negative answer needs a suitable counterexample.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 10.1.5.

Practice 10.1: Inclusion on sets of natural numbers

- **Reflexive.** $U \subseteq U$: an arbitrary element of U already belongs to U.
- **Not symmetric.** $\emptyset \subseteq \{0\}$, but $\{0\} \not\subseteq \emptyset$ because $0 \notin \emptyset$.
- **Antisymmetric.** If $U \subseteq V$ and $V \subseteq U$, then for every $n \in \mathbb{N}$, $n \in U \iff n \in V$. Set extensionality gives $U = V$.
- **Transitive.** If $U \subseteq V$ and $V \subseteq W$, take $n \in U$. The first inclusion gives $n \in V$, and the second gives $n \in W$. Thus $U \subseteq W$.

The chapter on sets already supplied the proof of antisymmetry; here we recognize its role as a relation property.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 10.1.5; set extensionality from Section 9.1.

Practice 10.1: Compare the coordinates separately

For $\preceq$ on $\mathbb{R}^2$:

- **Reflexive.** $x \leq x$ and $y \leq y$, so $(x, y) \preceq (x, y)$.
- **Not symmetric.** $(0, 0) \preceq (1, 1)$, but the reverse would require $1 \leq 0$.
- **Antisymmetric.** Relations in both directions give $x_1 \leq x_2 \leq x_1$ and $y_1 \leq y_2 \leq y_1$. Thus $x_1 = x_2$ and $y_1 = y_2$, so the pairs are equal.
- **Transitive.** From $(x_1, y_1) \preceq (x_2, y_2)$ and $(x_2, y_2) \preceq (x_3, y_3)$, obtain $x_1 \leq x_2 \leq x_3$ and $y_1 \leq y_2 \leq y_3$. Hence $(x_1, y_1) \preceq (x_3, y_3)$.

Not every two points are comparable: $(0, 1)$ and $(1, 0)$ provide an example.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 10.1.5.

10.2: Equivalence relations formalize a notion of sameness

Definition

A relation is an *equivalence relation* if it is reflexive, symmetric, and transitive.

Fix any $n \in \mathbb{Z}$. On $\mathbb{Z}$, let $x \sim y$ mean $x \equiv y \pmod{n}$, equivalently $y - x = nk$ for some $k \in \mathbb{Z}$.

- **Reflexive:** $x - x = n \cdot 0$.
- **Symmetric:** if $y - x = nk$, then $x - y = n(-k)$.
- **Transitive:** if $y - x = nk$ and $z - y = nl$, then $z - x = (z - y) + (y - x) = n(l + k)$.

The argument needs no positivity assumption on n. When $n = 0$, the relation is equality, because $0 \mid (y - x)$ holds exactly when $y - x = 0$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.2.1.

Equivalence classes collect the elements related to one element

For a relation on X and $a \in X$, define

$$[a] = \{b : X \mid a \sim b\}.$$

For congruence modulo 3 on $\mathbb{Z}$, there are three classes:

$$[0] = \{\ldots, -6, -3, 0, 3, 6, \ldots\}$$

$$[1] = \{\ldots, -5, -2, 1, 4, 7, \ldots\}$$

$$[2] = \{\ldots, -4, -1, 2, 5, 8, \ldots\}$$

Two integers are related exactly when they lie in the same row. Every integer is in its own class because the relation is reflexive.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 10.2.1 and 10.2.3.

Equal squares give another equivalence relation

On $\mathbb{Z}$, define $a \sim b$ by $a^2 = b^2$.

- **Reflexive:** $a^2 = a^2$ for every integer a.
- **Symmetric:** $a^2 = b^2$ implies $b^2 = a^2$.
- **Transitive:** $a^2 = b^2$ and $b^2 = c^2$ imply $a^2 = c^2$.

What are its classes? Since

$$a^2 = b^2 \quad \Longleftrightarrow \quad (a - b)(a + b) = 0,$$

we have $b = a$ or $b = -a$. Thus $[0] = \{0\}$ and $[a] = \{a, -a\}$ for $a \neq 0$.

This relation is not antisymmetric: $1 \sim -1$ and $-1 \sim 1$, yet $1 \neq -1$. Equivalence does not require antisymmetry.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.2.2.

Lean: Reuse the corresponding property of equality

```
def sameSquare (x y : Int) : Prop := x ^ 2 = y ^ 2

example : Reflexive sameSquare := by
  intro x
  rfl

example : Symmetric sameSquare := by
  intro x y h
  exact h.symm

example : Transitive sameSquare := by
  intro x y z hxy hyz
  exact hxy.trans hyz
```

Once the relation unfolds, reflexivity, symmetry, and transitivity become the corresponding facts about equality of squares.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.2.2; textbook-environment demonstration.

Related elements have equal classes

Theorem with its precise hypotheses

If $\sim$ is symmetric and transitive on X, and $a_1 \sim a_2$, then $[a_1] = [a_2]$.

Fix $b \in X$. We prove $a_1 \sim b \iff a_2 \sim b$.

- If $a_1 \sim b$, symmetry gives $a_2 \sim a_1$. Transitivity along $a_2 \sim a_1 \sim b$ gives $a_2 \sim b$.
- If $a_2 \sim b$, transitivity along $a_1 \sim a_2 \sim b$ gives $a_1 \sim b$.

Extensionality now gives the equality of sets. Reflexivity was not needed for this theorem.

Separately, if the relation is reflexive, then $a \in [a]$ for every a, because that membership statement is exactly $a \sim a$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.2.3.

Why equivalence classes form a partition

Assume $\sim$ is an equivalence relation on X.

- **The classes cover X.** Every $a \in X$ belongs to $[a]$ by reflexivity.
- **Overlapping classes are equal.** Suppose $c \in [a] \cap [b]$. Then $a \sim c$ and $b \sim c$. Symmetry gives $c \sim b$, and transitivity gives $a \sim b$. The preceding theorem yields $[a] = [b]$.

Thus any two classes are either equal or disjoint. The distinct classes split X into nonempty, nonoverlapping pieces: a *partition*.

For equality on $\mathbb{Z}$, each class is a singleton: $[a] = \{a\}$. For equal squares on $\mathbb{Z}$, most classes contain two integers instead.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.2.3 and Example 10.2.4; partition corollary derived here.

Pairs can represent the same rational value

On $\mathbb{Z} \times \mathbb{N}$, define

$$(a, b) \sim (c, d) \quad \iff \quad a(d+1) = c(b+1).$$

The products are taken in $\mathbb{Z}$; the natural numbers b, d are viewed as integers. Since $b+1, d+1 > 0$, this condition means

$$\frac{a}{b+1} = \frac{c}{d+1} \qquad \text{in } \mathbb{Q}.$$

For example, $(1, 1) \sim (2, 3)$ because $1 \cdot 4 = 2 \cdot 2$.

- Reflexivity follows from $a(b+1) = a(b+1)$.
- Symmetry follows by reversing the equality $a(d+1) = c(b+1)$.

Transitivity requires a calculation and cancellation of a nonzero factor.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.2.5.

Transitivity for the pair representation

Assume $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$. Set $B = b + 1$, $D = d + 1$, and $F = f + 1$, as integers. Then

$$aD = cB, \qquad cF = eD, \qquad D > 0.$$

We want $aF = eB$. First calculate

$$\begin{aligned} D(aF) &= (aD)F \\ &= (cB)F \\ &= (cF)B \\ &= (eD)B \\ &= D(eB). \end{aligned}$$

Since $D \neq 0$, cancellation gives $aF = eB$, as required. In the book's Lean proof, `ring` rearranges products, `rw` uses the hypotheses, and `cancel D at this` performs the final cancellation.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.2.5.

Having a bijection is an equivalence relation on types

For level-0 types X, Y (Lean: `Type`), define

$$X \sim Y \quad \Longleftrightarrow \quad \text{there exists a bijective function } f : X \to Y.$$

- **Reflexive.** The identity $\mathrm{id}_X : X \to X$ is its own inverse, so it is bijective.
- **Symmetric.** If $f : X \to Y$ is bijective, it has an inverse $g : Y \to X$. The two inverse identities also show that f is an inverse of g; hence g is bijective.
- **Transitive.** Given bijections $f : X \to Y$ and $g : Y \to Z$, their composition $g \circ f : X \to Z$ is injective and surjective by the composition proofs in Chapter 8.

The relation records that two types can be matched exactly, even when their elements look different.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.2.6; Section 8.3.

Practice 10.2: Prove equivalence and describe the classes

1. On $\mathbb{Z}$, define $a \sim b$ when there are positive integers m, n such that $am = bn$. Prove this is an equivalence relation and determine its classes.
2. On $\mathbb{N}^2$, define
$$(a, b) \approx (c, d) \quad \iff \quad a + d = b + c.$$
Prove this is an equivalence relation. Describe its classes using the integer difference $a - b$.

In the first problem, positivity is part of the definition. In the second, interpret differences in $\mathbb{Z}$, not as truncated subtraction in $\mathbb{N}$.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 10.2.7.

Practice 10.2: Positive multiples and the three sign classes

- **Reflexive:** use $m = n = 1$.
- **Symmetric:** if $am = bn$ with $m, n > 0$, reverse the equation and swap the witnesses.
- **Transitive:** suppose $am = bn$ and $bp = cq$, with all four witnesses positive. Then $a(mp) = b(np) = c(qn)$; both mp and qn are positive.

There are exactly three classes: the positive integers, $\{0\}$, and the negative integers.

- Positive multiplication preserves sign, so related integers have the same sign, including being zero together.
- If $a, b > 0$, choose $m = b, n = a$. If $a, b < 0$, choose $m = -b, n = -a$. These positive witnesses show that any two integers in either sign group are related.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 10.2.7, first problem.

Practice 10.2: Equal differences of natural-number pairs

For $(a, b), (c, d) \in \mathbb{N}^2$, the equation $a + d = b + c$ is equivalent to

$$a - b = c - d \qquad \text{when evaluated in } \mathbb{Z}.$$

- Reflexivity is $a + b = b + a$.
- Symmetry follows by reversing the equality of integer differences.
- Transitivity follows from $a - b = c - d$ and $c - d = e - f$, which give $a - b = e - f$, and hence $a + f = b + e$.

For each $k \in \mathbb{Z}$, one class is

$$C_k = \{(a, b) \in \mathbb{N}^2 \mid a - b = k \text{ in } \mathbb{Z}\}.$$

Each class is nonempty: use $(k, 0)$ if $k \geq 0$, or $(0, -k)$ if $k < 0$. Geometrically these classes lie on parallel diagonals in the natural-number grid.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 10.2.7, second problem.

Chapter 10 review: Classify the relation before using it

Relation and ambient type	Refl.	Symm.	Antisymm.	Trans.
$=$ on $\mathbb{R}$	yes	yes	yes	yes
$\mid$ on $\mathbb{N}$	yes	no	yes	yes
$(x-y)^2 \leq 1$ on $\mathbb{R}$	yes	yes	no	no
Loses-to on three hands	no	no	yes	no
$a^2 = b^2$ on $\mathbb{Z}$	yes	yes	no	yes

An equivalence relation needs the first, second, and fourth properties. Its equivalence classes cover the type, and overlapping classes are equal.

Exit check. If an equivalence relation is also antisymmetric, prove it is equality: $a \sim b$ gives $b \sim a$ by symmetry, hence $a = b$; reflexivity gives the converse.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 10.1.1–10.1.4; Section 10.2.

Reading and code context

Heather Macbeth, *The Mechanics of Proof*, Chapter 10:

- 10.1 Reflexive, symmetric, antisymmetric, transitive
- 10.2 Equivalence relations

https://hrmacbeth.github.io/math2001/10_Relations.html

These notes include original diagrams, worked practice solutions, and the derived partition corollary. Lean examples use the textbook's teaching environment; its relation-property names are `Reflexive`, `Symmetric`, `AntiSymmetric`, and `Transitive`. The spelling of antisymmetry follows that source environment.